# Practice Problems - Midterm.

Q1. A stock is priced at $S_0 = 10$. In one period, it can go up by a factor $u = 1.\overline{5}$, or down by a factor $d = 0.\overline{5}$. The continuously compounded risk-free rate is $r = 0.1$ per period. Consider a European put option with strike $K = 10$ maturing after one step.

a) Compute the risk-neutral probability $p^*$.

Solution: $p^* = \frac{e^r - d}{u - d} = \frac{e^{0.1} - 0.\overline{5}}{1.\overline{5} - 0.\overline{5}} \approx 0.60\overline{5}$.

b) Find the option payoff at maturity in both up and down states.

Solution: $S_{up} = S_0 \cdot u = 10 \times 1.\overline{5} = 1\overline{5}$

payoff $= \max(10 - 15, 0) = 0$

$S_{down} = S_0 \cdot d = 10 \times 0.\overline{5} = \overline{5}$, payoff $= \max(10 - 5, 0) = 5$

Q2. Suppose $Z(t)$ is a standard Brownian motion, where increments are defined as

$$dZ = \phi\sqrt{\Delta t}, \quad \phi \sim N(0,1).$$

Define a new process $Y(t) = Z(t) - t$.

a) Compute the expectation $E[Y(t)]$.

Solution: Expectation

$$E[Y(t)] = E[Z(t) - t] = \underbrace{E[Z(t)]}_{0} - t = -t$$

b) Compute the variance $Var[Y(t)]$.

Solution:

$$Var[Y(t)] = E[Y(t)^2] - (E[Y(t)]^2)$$

Now

$$Y(t)^2 = (Z(t) - t)^2 = Z(t)^2 - 2tZ(t) + t^2$$

Take expectation:

$$E[Y(t)^2] = \underbrace{E[Z(t)^2]}_{E[t\phi^2] = tE[\phi^2] = t.} - 2t\overbrace{E[Z(t)]}^{0} + t^2$$

$$E[Y(t)^2] = t + t^2$$

$$\Longrightarrow \quad \text{Var}[Y(t)] = (t + t^2) - (-t)^2 = t \quad \checkmark$$

Note. There are various ways to solve this.

Q3. Suppose a stock price follows geometric Brownian motion: $dS = \mu S dt + \sigma S dZ$

Let $Y(t) = \sqrt{S(t)}$.

a) Apply Ito's Lemma to drive the SDE for $Y(t)$.

Solution: We take $f(S) = \sqrt{S} = S^{1/2}$

$$\frac{\partial f}{\partial S} = \frac{1}{2} S^{-1/2}, \qquad \frac{\partial^2 f}{\partial S^2} = \frac{-1}{4} S^{-3/2}$$

By Ito:

$$dY = \left(\mu S \cdot \frac{1}{2} S^{-1/2} + \frac{1}{2} \sigma^2 S^2 \left(\frac{-1}{4} S^{-3/2}\right)\right) dt + \sigma S \left(\frac{1}{2} S^{-1/2}\right) dZ$$

$$= \left(\frac{\mu}{2} - \frac{\sigma^2}{8}\right) \sqrt{S}\, dt + \frac{\sigma}{2} \sqrt{S}\, dZ$$

b) Identify the drift term and the random term.

Solution: $\text{drift} = \left(\frac{\mu}{2} - \frac{\sigma^2}{8}\right) \sqrt{S}$

$$\text{random term} = \frac{\sigma}{2} \sqrt{S}$$

Q4. You are offered a game with 2 stages:

- Stage 1: observe a reward $R_1$ uniformly distributed on $[0,2]$. You may accept it and end, or reject it and move to Stage 2.
- Stage 2: Reward $R_2$ is uniformly distributed on $[0,3]$. You must accept it.

Compute the expected value of playing optimally.

Solution: At stage 2: $E[R_2] = \frac{0+3}{2} = 1.5$

At stage 1: if $R_1 \geq 1.5$, accept, else reject.

Expected Value $= E[R_1 \mid R_1 \geq 1.5] \cdot P(R_1 \geq 1.5)$

$+ E[R_2] \cdot P(R_1 < 1.5)$

Now range of $R_1 : [0,2]$. We accept only if $R_1 \in [1.5, 2]$. Now probability of this happening:

$P(R_1 \geq 1.5) = \frac{0.5}{2} = 0.25$

Also $E[R_1 \mid R_1 \geq 1.5] = 1.5 + 2 / 2 = 1.75$

$\Longrightarrow$ $1.75 \times 0.25 = 0.4375$

Now for the case of rejecting $R_1$ if it's small:

$P(R_1 < 1.5) = \frac{1.5}{2} = 0.75$

Also, if we reject, we go to stage 2,

$E[R_2] = 1.5$

$\Longrightarrow 1.5 \times 0.75 = 1.125$

$\Longrightarrow$ Total: $0.4375 + 1.125 = 1.5625$